

Question Number	Answer	Additional guidance	Mark
4(a)	A calculation showing the following steps: • mass of bran calculated (1) • mass of fibre calculated (1) • correct answer to two significant figures (1)	<u>Example of calculation</u> $(48 \div 100) \times 14 = 6.72 \text{ (mg)}$ $(6.72 \div 100) \times 43 = 2.889 \text{ (mg)}$ 2.9 mg Correct answer with no working scores full marks	(3)

Question Number	Answer	Additional guidance	Mark
4(b)	An explanation that includes the following points: • more sustainable (than oil-based plastic) (1) • biodegradable / can be broken down by decomposers (1) • carbon neutral (1)	ACCEPT ORA for oil-based plastic ACCEPT renewable / can be regrown / available for future generations ALLOW can decompose ACCEPT {does not contribute to / reduces} {greenhouse effect / global warming}	(3)

Question Number	Answer	Additional guidance	Mark
4(c)	An explanation that includes three of the following points: • pollen tube transports {generative nucleus/male nuclei} to {embryo sac / ovary / micropyle} (1) • (by releasing) digestive enzymes (1) • one of the male nuclei {fertilises / fuses with} the (two) polar nuclei (1) • (causing) the formation of a {<u>3n/ triploid</u>} endosperm (nucleus) (1)	ACCEPT pollen tube transports {generative nucleus/male nuclei} down the style ACCEPT male gamete / sperm nucleus ACCEPT male gamete / {sperm / haploid} nucleus	(3)